

DBMS LAB1

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AIM: To perform and understand fundamental DBMS operations, including database creation , table management, data manipulation, modification, and retrieval using MySQL.

```
MySQL Shell 8.0.46

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Type '\help' or '\?' for help; '\quit' to exit.
MySQL JS > \sql
Switching to SQL mode... Commands end with ;
MySQL SQL > \connect root@localhost:3306
Creating a session to 'root@localhost:3306'
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 15
Server version: 8.0.46 MySQL Community Server - GPL
No default schema selected; type \use <schema> to set one.
MySQL localhost:3306 ssl SQL > SHOW databases;
+-----+
| Database |
+-----+
| anmol_68 |
| information_schema |
| janhavi_26 |
| mahek_05 |
| mitsu_d6 |
| mysql |
| performance_schema |
| stud1_match |
| stud1_test |
| stud1test |
| sys |
| yadav_65 |
+-----+
```

```
MySQL localhost:3306 ssl SQL > CREATE DATABASE keswani_11;
Query OK, 1 row affected (0.0049 sec)
MySQL localhost:3306 ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| anmol_68 |
| information_schema |
| janhavi_26 |
| keswani_11 |
| mahek_05 |
| mitsu_d6 |
| mysql |
| performance_schema |
| stud1_match |
| stud1_test |
| stud1test |
| sys |
| yadav_65 |
+-----+
13 rows in set (0.0009 sec)
MySQL localhost:3306 ssl SQL > CREATE DATABASE dnya_05;
Query OK, 1 row affected (0.0065 sec)
MySQL localhost:3306 ssl SQL > SHOW DATABASES;
```

```

+-----+
| Database |
+-----+
| anmol_68 |
| dnya_05  |
| information_schema |
| janhavi_26 |
| keswani_11 |
| mahek_05  |
| mitsu_d6  |
| mysql     |
| performance_schema |
| stud1_match |
| stud1_test |
| stud1test |
| sys       |
| yadav_65  |
+-----+
14 rows in set (0.0010 sec)
MySQL localhost:3306 ssl SQL > DROP DATABASE mitsu_d6;
Query OK, 1 row affected (0.0119 sec)

```

```

MySQL localhost:3306 ssl SQL > SHOW databases;
+-----+
| Database |
+-----+
| anmol_68 |
| dnya_05  |
| information_schema |
| janhavi_26 |
| keswani_11 |
| mahek_05  |
| mysql     |
| performance_schema |
| stud1_match |
| stud1_test |
| stud1test |
| sys       |
| yadav_65  |
+-----+
13 rows in set (0.0010 sec)
MySQL localhost:3306 ssl SQL > USE yadav_65;
Default schema set to 'yadav_65'.
Fetching global names, object names from 'yadav_65' for auto-completion... Press ^C to stop.
MySQL localhost:3306 ssl yadav_65 SQL > CREATE TABLE Persons (
    -> PersonID int,
    -> LastName varchar(25),
    -> FirstName varchar(25),
    -> Address varchar(100),
    -> City varchar(25)
    -> );
Query OK, 0 rows affected (0.0101 sec)
MySQL localhost:3306 ssl yadav_65 SQL > SHOW TABLES;
+-----+

```

```

+-----+
| Tables_in_yadav_65 |
+-----+
| persons |
+-----+
1 row in set (0.0011 sec)
MySQL localhost:3306 ssl yadav_65 SQL > DESCRIBE Persons;

```

Field	Type	Null	Key	Default	Extra
PersonID	int	YES		NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	

```

5 rows in set (0.0019 sec)
MySQL localhost:3306 ssl yadav_65 SQL > DROP TABLE Persons;
Query OK, 0 rows affected (0.0103 sec)

```

```

MySQL localhost:3306 ssl yadav_65 SQL > CREATE TABLE Customers (CustomerID int,LastName varchar(25),FirstName
-> varchar(25),Address varchar(100),City varchar(25), primary key
-> (CustomerID));
Query OK, 0 rows affected (0.0144 sec)

```

```

MySQL localhost:3306 ssl yadav_65 SQL > DESCRIBE Customers;

```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	

```

5 rows in set (0.0020 sec)
MySQL localhost:3306 ssl yadav_65 SQL > INSERT INTO Customers (CustomerID,LastName,FirstName,Address,City)
-> VALUES (10,'yadav','khushi','A67,Building10,abc lane','Mumbai');
Query OK, 1 row affected (0.0048 sec)

```

```

MySQL localhost:3306 ssl yadav_65 SQL > SELECT * FROM Customers;

```

CustomerID	LastName	FirstName	Address	City
10	yadav	khushi	A67,Building10,abc lane	Mumbai
96365	Shah	Hiren	A67,Building10,abc lane	Mumbai

```

2 rows in set (0.0006 sec)

```

```

MySQL localhost:3306 ssl yadav_65 SQL > insert into customers values
-> (300,'keswani','anmol','bk16 room 1','mumbai'),
-> (400,'yadav','janhavi','jaoli apt','mumbai'),
-> (500,'yadav','khushi','mayflower apt','pune'),
-> (600,'punjabi','kani','mayflower','mumbai'),
-> (700,'kalwani','palak','satnam apt','mumbai'),
-> (800,'bhatnagar','mahek','tulip-b nisarg','mumbai'),
-> (900,'sadwani','vivaan','lily-b nisarg','ulhasnagar'),
-> (901,'tharyani','mayank','shiv pooja','ulhasnagar'),
-> (902,'bhatnagar','ria','shiv pooja','mumbai'),
-> (903,'mankani','dhiren','nisarg greens','mumbai');
Query OK, 10 rows affected (0.0043 sec)

```

```

Records: 10 Duplicates: 0 Warnings: 0

```

```

MySQL localhost:3306 ssl yadav_65 SQL > select *from customers;

```

CustomerID	LastName	FirstName	Address	City
10	yadav	khushi	A67,Building10,abc lane	Mumbai
300	keswani	anmol	bk16 room 1	mumbai
400	yadav	janhavi	jaoli apt	mumbai
500	yadav	khushi	mayflower apt	pune
600	punjabi	kani	mayflower	mumbai
700	kalwani	palak	satnam apt	mumbai
800	bhatnagar	mahek	tulip-b nisarg	mumbai
900	sadwani	vivaan	lily-b nisarg	ulhasnagar
901	tharyani	mayank	shiv pooja	ulhasnagar
902	bhatnagar	ria	shiv pooja	mumbai
903	mankani	dhiren	nisarg greens	mumbai
96365	Shah	Hiren	A67,Building10,abc lane	Mumbai

```

12 rows in set (0.0006 sec)

```

```
MySQL localhost:3306 ssl yadav_65 SQL > alter table customers add email varchar(50);
Query OK, 0 rows affected (0.0148 sec)
```

Records: 0 Duplicates: 0 Warnings: 0

```
MySQL localhost:3306 ssl yadav_65 SQL > describe customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	
email	varchar(50)	YES		NULL	

6 rows in set (0.0016 sec)

```
MySQL localhost:3306 ssl yadav_65 SQL > alter table customers add remark varchar(255);
Query OK, 0 rows affected (0.0119 sec)
```

Records: 0 Duplicates: 0 Warnings: 0

```
MySQL localhost:3306 ssl yadav_65 SQL > describe customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	
email	varchar(50)	YES		NULL	
remark	varchar(255)	YES		NULL	

7 rows in set (0.0015 sec)

```
MySQL localhost:3306 ssl yadav_65 SQL > select *from customers;
```

CustomerID	LastName	FirstName	Address	City	email	remark
10	yadav	khushi	A67,Building10,abc lane	Mumbai	NULL	NULL
300	keswani	anmol	bk16 room 1	mumbai	NULL	NULL
400	yadav	janhavi	jaoli apt	mumbai	NULL	NULL
500	yadav	khushi	mayflower apt	pune	NULL	NULL
600	punjabi	kani	mayflower	mumbai	NULL	NULL
700	kalwani	palak	satnam apt	mumbai	NULL	NULL
800	bhatnagar	mahek	tulip-b nisarg	mumbai	NULL	NULL
900	sadwani	vivaan	lily-b nisarg	ulhasnagar	NULL	NULL
901	tharyani	mayank	shiv pooja	ulhasnagar	NULL	NULL
902	bhatnagar	ria	shiv pooja	mumbai	NULL	NULL
903	mankani	dhiren	nisarg greens	mumbai	NULL	NULL
96365	Shah	Hiren	A67,Building10,abc lane	Mumbai	NULL	NULL

12 rows in set (0.0006 sec)

```
MySQL localhost:3306 ssl yadav_65 SQL > update customers
-> set email='ramesh.joshi@abc.com',
-> remark='regular buyer'
-> where customerID=89790;
```

Query OK, 0 rows affected (0.0083 sec)

Rows matched: 0 Changed: 0 Warnings: 0

```
MySQL localhost:3306 ssl yadav_65 SQL > SELECT CustomerID,Address FROM Customers;
```

CustomerID	Address
10	A67,Building10,abc lane
300	bk16 room 1
400	jaoli apt
500	mayflower apt
600	mayflower
700	satnam apt
800	tulip-b nisarg
900	lily-b nisarg
901	shiv pooja
902	shiv pooja
903	nisarg greens
96365	A67,Building10,abc lane

12 rows in set (0.0006 sec)

```
MySQL localhost:3306 ssl yadav_65 SQL > |
```

1. What is a Database? Why do we need a Database Management System such as MySQL?

A **database** is an organized collection of related data that allows information to be stored, managed, and retrieved efficiently. It helps users store large amounts of data in a structured manner.

A **Database Management System (DBMS)** is software used to create, store, organize, retrieve, and manage data in a database. **MySQL** is a popular relational DBMS that uses **SQL (Structured Query Language)** to work with data.

We need a DBMS such as MySQL because it:

- Stores data systematically and reduces unnecessary duplication.
- Makes data retrieval faster and easier using SQL queries.
- Allows data to be inserted, updated, and deleted efficiently.
- Provides security by controlling access to the database.
- Maintains data integrity and consistency.
- Handles large amounts of data efficiently.
- Supports multiple users working with the database.

In conclusion, a database stores organized information, while a DBMS such as MySQL provides the tools required to manage that information efficiently, securely, and reliably.

2. What is the Difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE

- **CREATE DATABASE** — This command is used to **create a new database** in MySQL. It provides a separate space where related tables and data can be stored and managed.

Example : CREATE DATABASE college;

- **SHOW DATABASES** — This command is used to **display the list of all databases** currently available on the MySQL server. It helps the user view the databases that have been created.

Example : SHOW DATABASES;

- **DROP DATABASE** — This command is used to **permanently delete an existing database** from the MySQL server, including all the tables and data stored within it.

Example : DROP DATABASE college;

3. What is a Table in a Database? Explain the Purpose of Rows and Columns.

A **table** in a database is a structured way of storing related data in the form of **rows and columns**. It organizes information systematically, making it easier to store, retrieve, update, and manage data. A table generally represents a particular entity, such as students, employees, or products.

- **Rows** — A row represents a **single record or entry** in a table. Each row contains complete information about one particular entity. For example, in a student table, one row can contain the details of one student.
- **Columns** — A column represents a **specific attribute or field** of the records. It defines what type of information is stored in that column, such as student ID, name, age, or department.

Therefore, **rows represent individual records, while columns represent the attributes or fields of those records.** Together, they provide a structured and organized representation of data in a database.

4. What is a Primary Key? Why was CustomerID selected as the primary key in the Customers table?

A **primary key** is a field or a combination of fields used to uniquely identify each record in a database table. Each value of a primary key must be unique, and it cannot contain a NULL value. It helps in identifying individual records and maintaining the integrity and consistency of the data.

CustomerID was selected as the primary key in the Customers table because it provides a unique identity to every customer. Since each customer is assigned a different CustomerID, two customers cannot have the same identifier. It also allows a particular customer's record to be easily identified, accessed, and referenced.

Therefore, CustomerID is suitable as a primary key because it uniquely identifies each customer and prevents duplicate or unidentified customer records.

5. What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

- **VARCHAR** is a data type used to store **variable-length text or character data**. It is suitable for storing information such as names, addresses, email addresses, and other textual values. The number specified in VARCHAR(n) represents the maximum number of characters that can be stored.

Example: CustomerName VARCHAR(50) — stores the name of a customer, with a maximum limit of 50 characters.

- **INT** is a data type used to store **whole numbers or integers**. It is commonly used for numerical values such as IDs, age, quantity, marks, or counts. It does not store decimal values.

Example: CustomerID INT — stores the unique numerical identification number of a customer.

Therefore, **VARCHAR is used for variable-length text data, while INT is used for whole-number data.**

6. What is the Difference between INSERT, SELECT, UPDATE, and ALTER TABLE

- **INSERT** — This command is used to **add new records or rows** to an existing table. It is used when new data needs to be stored in the database.

Example : INSERT INTO Customers VALUES (101, 'Rahul', 'Mumbai');

- **SELECT** — This command is used to **retrieve or display data** from one or more tables. It allows users to view the required records and information stored in the database.

Example : SELECT * FROM Customers;

- **UPDATE** — This command is used to **modify or change existing data** in a table. It is useful when the information stored in one or more records needs to be corrected or updated.

Example : UPDATE Customers SET City = 'Pune' WHERE CustomerID = 101;

- **ALTER TABLE** — This command is used to **modify the structure of an existing table**. It can be used to add, modify, or remove columns and make other structural changes.

Example : ALTER TABLE Customers ADD Email VARCHAR(50);

In short: INSERT adds data, SELECT retrieves data, UPDATE modifies data, while ALTER TABLE modifies the structure of a table.

7. Why is the WHERE clause important in an UPDATE statement?

The **WHERE clause** is used to specify **which particular record or records should be updated**. It ensures that only the intended rows are modified based on the given condition.

For example:

```
UPDATE Customers
```

```
SET City = 'Pune'
```

```
WHERE CustomerID = 101;
```

Here, only the customer whose CustomerID is **101** will be updated.

If we write an UPDATE statement **without a WHERE condition**, the specified column will be updated for **all the records in the table**. This can unintentionally change the data of every row and may result in **loss or incorrect modification of important information**.

Example:

```
UPDATE Customers
```

```
SET City = 'Pune';
```

This will change the city to **Pune for every customer** in the table.

Therefore, the WHERE clause is important because it **prevents unintended changes and ensures that only the required records are updated**.

